Limitations and future perspectives for satellite-based soil carbon monitoring

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ABSTRACT ORIGINAL

Soil organic carbon (SOC) plays a crucial role in terrestrial C storage and ecosystem services. Agricultural management practices have the potential to increase C inputs and reduce its losses. However, uniform standard protocols for measuring, monitoring, and assessing changes using remote sensing is lacking for SOC in the scientific literature. In this discussion paper, we present techniques for collecting and analyzing ground samples and employing remote sensing to quantify SOC, along with its limitations and future perspectives. Our analysis identified a number of key limitations to advancing the science for remotely sensed terrestrial C in croplands including i) lack of consensus in sampling depth and density, ii) the absence of a standard (or universally accepted) laboratory procedure and statistical methodology, and iii) lack of details on imagery pre-processing or information on the spectral properties of the targeted soils. Establishing standard protocols for ground-truth data collection and remote sensing approaches, as well as a knowledge of the impacts of diverse soil types, land uses, and landscapes on C assessment, are all required to enhance the accuracy and reliability of future SOC assessments.